

Unseen Seminar Questions 1: Answers

Probability and Statistical Inference

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1. **Assessing a study design.** A company recruits 120 volunteers from one of its depots to study a new shift reminder. Within this volunteer sample, the reminder is assigned at random. The target of interest is the ATE for all of the company's night-shift workers. Mean sleep is 6.8 hours in the assigned-reminder group and 5.9 hours in the control group.
- (a) State a natural estimator and its numerical estimate.
 - (b) For which population does random assignment allow this comparison to identify a causal effect?
 - (c) Why does the experiment not automatically identify the company-wide ATE? Would collecting many more volunteers from the same depot necessarily solve this generalization problem?

Solution.

- (a) A natural estimator is the difference between the two sample means,

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0.$$

This is a random rule before random assignment and outcome observation. In the realized study its estimate is

$$\hat{\beta} = 6.8 - 5.9 = 0.9 \text{ hours.}$$

- (b) Random assignment makes reminder status unrelated to the study participants' potential outcomes. Therefore, the treated-control difference in means estimates the ATE for the study participants. The value 0.9 is still a sample estimate. Because the random assignment could have produced different groups, this estimate can differ from the participants' exact ATE.
 - (c) The participants volunteered and all came from one depot. They need not represent night-shift workers at other depots, so the participant effect need not equal the company-wide ATE. More volunteers selected in the same way can reduce sampling variation for the comparison at this depot. However, a larger sample of these volunteers does not show that they represent the company's full workforce. Random assignment makes treatment groups comparable within the study; it does not randomly sample workers from the company-wide population.
2. **Bias, variance and MSE.** Two estimators target $\theta = 3$. Across repeated samples, T takes values 1, 3, 5 with probabilities $1/4, 1/2, 1/4$, whereas S always equals 3.5. You are given that T has sampling mean 3 and variance 2.
- (a) Calculate the bias, variance and MSE of each estimator.
 - (b) Which has lower MSE? Why does this example not show that a constant estimator is always a sensible choice?

Solution.

- (a) Using the supplied mean and variance,

$$\text{Bias}(T) = 3 - 3 = 0, \quad \text{Var}(T) = 2, \quad \text{MSE}(T) = 0^2 + 2 = 2.$$

S always equals 3.5, so its sampling mean is 3.5 and its variance is zero. Hence

$$\text{Bias}(S) = 0.5, \quad \text{Var}(S) = 0, \quad \text{MSE}(S) = 0.5^2 + 0 = 0.25.$$

- (b) S has lower MSE because $0.25 < 2$. The fixed guess 3.5 happens to be close to the target 3. For another target, it could have a large squared bias. A constant ignores the data and does not improve as the sample grows.

3. **A short Bayesian update.** A part comes from machine A with probability 0.40 and from machine B otherwise. The probability that a part is red is 0.75 under A and 0.25 under B . A red part is observed. Find the posterior probability that it came from A . Label the prior, likelihood and normalizing probability in your calculation.

Solution. Let R denote a red part. The prior is $\mathbb{P}(A) = 0.40$, and the likelihood of the observation under A is $\mathbb{P}(R | A) = 0.75$. The normalizing probability is

$$\begin{aligned} \mathbb{P}(R) &= \mathbb{P}(R | A)\mathbb{P}(A) + \mathbb{P}(R | B)\mathbb{P}(B) \\ &= 0.75(0.40) + 0.25(0.60) = 0.30 + 0.15 = 0.45. \end{aligned}$$

Thus Bayes' rule gives

$$\mathbb{P}(A | R) = \frac{\mathbb{P}(R | A)\mathbb{P}(A)}{\mathbb{P}(R)} = \frac{0.30}{0.45} = \frac{2}{3}.$$

The observation raises the probability of A from the prior 0.40 to the posterior $2/3$ because red is more likely under A than under B .

4. **Sample quantiles.** For the ordered sample

$$3, 3, 4, 4, 5, 7, 7, 9,$$

use $\hat{q}_p = X_{(\lceil np \rceil)}$ for all reported quantiles.

- (a) Compute the mean, $\hat{q}_{.25}$, $\hat{q}_{.50}$, $\hat{q}_{.75}$ and the IQR.
 (b) Replace 9 by 41. Which of these summaries change, and by how much?

Solution.

- (a) The sum is 42, so the mean is $42/8 = 5.25$. Under the specified convention,

$$\hat{q}_{.25} = X_{(\lceil 2 \rceil)} = X_{(2)} = 3, \quad \hat{q}_{.50} = X_{(\lceil 4 \rceil)} = X_{(4)} = 4, \quad \hat{q}_{.75} = X_{(\lceil 6 \rceil)} = X_{(6)} = 7.$$

The IQR is $7 - 3 = 4$. (A software convention that averages the two central order statistics would report a different median; the question deliberately specifies the lecture's order-statistic convention.)

- (b) The new sum is $42 - 9 + 41 = 74$, so the mean becomes $74/8 = 9.25$, an increase of 4. The replacement changes only the largest order statistic. The second, fourth and sixth order statistics are still 3, 4, 7, so all three reported quantiles and the IQR remain unchanged.

5. **True or false?** State whether each claim is true or false and give a one- or two-sentence explanation.

- (a) The condition $D \perp\!\!\!\perp Y(0)$ is enough for a treated-control population mean difference to identify the ATET, but it is not by itself enough to identify the ATE.

- (b) Even if the target θ can vary continuously across populations, an estimator's sampling distribution at a fixed sample size can be discrete.
- (c) Two estimators with the same bias and variance must have the same sampling distribution.
- (d) A causal parameter can be identified even when the available estimator has high sampling variance.
- (e) Consistency does not require an estimator to have a normal sampling distribution at any finite sample size.

Solution.

- (a) **True.** The decomposition

$$\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0] = \text{ATE} + \mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0]$$

has a zero final term under $D \perp\!\!\!\perp Y(0)$. To identify the ATE, we also need enough comparability in $Y(1)$. For the mean ATE, the required additional restriction can be stated as

$$\mathbb{E}[Y(1) \mid D = 1] = \mathbb{E}[Y(1) \mid D = 0].$$

The full conditional distributions of $Y(1)$ do not have to be equal. Independence from $Y(0)$ alone does not imply even the displayed equality of means.

- (b) **True.** For example, at fixed n a sample proportion based on zero-or-one observations takes only the values $0, 1/n, \dots, 1$, even though the population probability it targets can take any value in $[0, 1]$.
- (c) **False.** Bias and variance describe only a distribution's centre and spread. Distinct sampling distributions can share the same mean and variance; for example, suitably scaled normal and two-point errors can both have mean zero and variance one.
- (d) **True.** Identification asks whether the population distribution of the observed data determines the causal target. Sampling variance asks how much a particular estimator changes across samples. It can be large even when the target is identified.
- (e) **True.** Consistency requires concentration at the target, not a particular finite-sample shape. For example, if U equals -1 or 1 with equal probability, $\hat{\theta}_n = \theta + U/n$ is two-point distributed for every n but converges in probability to θ .